SCISOFT ACADEMY

OBJECT ORIENTED PROGRAMMING

EPISODE - 8

ABSTRACTION

WHAT vs HOW

DRIVING A CAR:

You use:

→ Steering
→ Accelerator
→ Brake

Do you need to understand:

→ Engine combustion ?
→ Fuel injection ?
→ ECU algorithms ?
→ Transmission internals ?

NO

DRIVER
↓
[Simple Interface]
Accelerator | Brake | Steering
↓
Internal Complexity

Abstraction
↓
Expose what is necessary
+
Hide unnecessary implementation complexity

WHAT
↓
Interface

How
↓
Implementation

Example:

>> payment.pay(₹1,000)

User sees ⇒ Just "Pay".

Internally,

Pay
↓
Validate
↓
Authorize
↓
Process
↓
Confirm

✓ Caller need not manage every internal step.

Encapsulation vs Abstraction

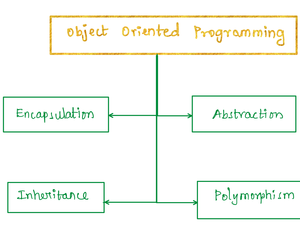
Encapsulation:

Focus:
Controlling access to state and bundling behaviour.

Abstraction:

Focus:
→ Expose essential operations while hiding unnecessary complexity.

* They are related, but not same concept.



NEXT VIDEO

Interface & Abstract Clases ...